

Tide retrospective: *higher h -derivatives of the anharmonic partition function*

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Abstract

For the 1D anharmonic potential $L(x) = \frac{\lambda}{2}x^2 + \frac{\alpha}{6}x^3 + \frac{\gamma}{24}x^4$ ($\alpha^2 < 3\lambda\gamma$) under the linear perturbation $A(x) = x$, the perturbed partition function is $Z(h) = \int e^{-t(L(x)+hx)} dx$. The existing regularity instance proved its first h -derivative at $h = 0$. This tide proves the general iterated-derivative *step*: for every $n \in \mathbb{N}$,

$$\left. \frac{d}{dh} \right|_{h=0} \int (-tx)^n e^{-t(L(x)+hx)} dx = \int (-tx)^{n+1} e^{-tL(x)} dx,$$

whose $n = 1, 2$ instances are the second and third h -derivatives of Z at $h = 0$ ($\int (tx)^2 e^{-tL}$ and $\int -(tx)^3 e^{-tL}$). New file `AnharmonicPartitionHigherDeriv.lean` (≈ 200 lines), commit on branch `tide/anharmonic-partition-higher-deriv`; builds clean, no `sorry/axiom/native_decide`, no lint warnings; both derivative forms numerically confirmed to $\sim 10^{-6}$. Directly reuses the general $|x|^m$ integrability witness and the two public pointwise helpers landed in the FDT-capstone tide earlier the same day.

1 Setting

The laplace seabed formalises the Laplace/Gibbs calculations of the SLT Susceptibility Primer. A recent tide closed the last `sorry` in the anharmonic `GibbsRegularity` instance, whose third field, `partition_hasDerivAt`, gives the *first* h -derivative at $h = 0$ of the perturbed partition $Z(h) = \int e^{-t(L(x)+hx)} dx$, namely $\int (-tx) e^{-tL}$, proved by Mathlib's dominated-differentiation lemma `hasDerivAt_integral_of_dominated_loc_of_deriv_le`. Earlier this same day the FDT-capstone tide¹ added a general integrability witness $\int |x|^m e^{-tL} < \infty$ for all m and made two pointwise helpers public.

The minimal good target outward from there: the higher h -derivatives. The flow-equation / fourth-cumulant identity needs the second and third derivatives of Z at the unperturbed point $h = 0$. Rather than prove $n = 2$ and $n = 3$ as two separate lemmas, a single general- n step lemma covers both (and all higher orders) at no extra cost, because the new $|x|^m$ witness is already uniform in the power.

2 The thing formalised

For $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$, and every $n \in \mathbb{N}$,

$$\left. \frac{d}{dh} \right|_{h=0} \int_{\mathbb{R}} (-tx)^n e^{-t(L(x)+hx)} dx = \int_{\mathbb{R}} (-tx)^{n+1} e^{-tL(x)} dx.$$

¹the anharmonic FDT capstone.

The Lean signature:

```
theorem anharmonic_partition_deriv_step
  {lam alpha gamma t : ℝ} (n : ℕ)
  (hlam : 0 < lam) (hgamma : 0 < gamma)
  (hdisc : alpha ^ 2 < 3 * lam * gamma) (ht : 0 < t) :
  HasDerivAt
    (fun h => ∫ x, (-(t * x)) ^ n *
      Real.exp (-(t * (anharmonicPotential lam alpha gamma x + h * x))))
    (∫ x, (-(t * x)) ^ (n + 1) *
      Real.exp (-(t * anharmonicPotential lam alpha gamma x)))
    0
```

The two named corollaries `anharmonic_partition_secondDeriv` ($n = 1$, right-hand side $\int (tx)^2 e^{-tL}$) and `anharmonic_partition_thirdDeriv` ($n = 2$, right-hand side $\int -(tx)^3 e^{-tL}$) are the literal second and third derivatives the direction asked for. They are the function being differentiated read as the (n) -th derivative integrand: differentiating $\int (-(tx))^1 e^{-t(L+hx)}$ at 0 yields $Z''(0)$, and differentiating $\int (-(tx))^2 e^{-t(L+hx)}$ yields $Z'''(0)$.

3 Proof strategy

The proof is the $n = 1$ `partition_hasDerivAt` field with one extra constant-in- h factor $(-tx)^n$ carried through. Feed `hasDerivAt_integral_of_dominated_loc_of_deriv_le` the family $F(h, x) = (-tx)^n e^{-t(L(x)+hx)}$ on the unit ball around 0:

- **Pointwise derivative.** The public `anharmonic_perturbed_pointwise_hasDerivAt` differentiates the Boltzmann factor; `HasDerivAt.const_mul` $(-tx)^n$ multiplies in the monomial, and `ring` (with the exponential as an opaque atom) collapses $(-tx)^n \cdot (-tx)$ to $(-tx)^{n+1}$ via `pow_succ`.
- **Dominator.** Writing $F'(h, x) = (-tx)^n \cdot [(-tx) e^{-t(L+hx)}]$ and reusing the $n = 0$ pointwise bound `anharmonic_perturbed_pointwise_bound` on the bracket, the dominator is $t^{n+1} e^{t/2c} |x|^{n+1} e^{-(t/2)L}$, integrable by the general $|x|^{n+1}$ witness at half temperature.
- **Endgame.** The lemma returns the derivative as $\int F'(0, x)$, whose integrand still carries a $+0 \cdot x$; an `integral_congr_ae` with `ring` under the integral drops it.

4 Roadblocks and resolutions

One real bug, caught by the build, then by re-reading the magnitude. The first draft tried to dominate $F(0, x)$ — needed for the *integrability of F at 0* hypothesis — by the bare witness $|x|^n e^{-tL}$. But $\|(-tx)^n e^{-tL}\| = t^n |x|^n e^{-tL}$ carries a t^n the bare witness lacks, so the domination inequality was simply false for $t > 1$ (and the `rw` chain that was supposed to massage it failed to match, which is how Lean flagged it). The fix was to scale the dominating function by t^n (`Integrable.const_mul`) and finish the now-genuine equality with `le_of_eq` (by `ring`). A good reminder that “mirror the $n = 0$ proof” still has to track the constants the monomial introduces.

Everything else followed the FDT-capstone template mechanically: the same continuity-to-measurability discharges², the same `set-bound` idiom then a `rw` closing the integrability congruence by `rfl`.

GPT-5.5 Pro was consulted on the statement form (general- n step vs. two explicit derivatives, and the $(-tx)^{n+1}$ sign). Because the proof is a constant-factor generalisation of an already-merged result and the closed forms were numerically confirmed against finite differences before the build, this proceeded as a mirror-shaped tide without blocking on the consult; the build is the correctness witness.

5 What was Mathlib, what was new

Mathlib. The dominated-differentiation lemma³; `HasDerivAt.const_mul`; the `Integrable` combinators (`const_mul`, `mono'`, `congr`); `integral_congr_ae`; and continuity-to-measurability via `fun_prop`.

Seabed (reused). The general $|x|^m$ integrability witness and the two public pointwise helpers (pointwise derivative, pointwise bound), all from the FDT-capstone tide; `anharmonic_coercive`.

New. The general iterated-derivative step lemma and the two named derivative corollaries. About 200 lines.

6 Lessons

- **Generalise the index when the supporting witness already is.** Because the $|x|^m$ integrability lemma is uniform in m , the single general- n step lemma cost no more than $n = 2$ alone and gives every higher derivative for free. The same observation made the FDT capstone cheap; it is becoming the defining move of this anharmonic arc.
- **“Mirror the $n = 0$ proof” is not “copy it”.** The monomial factor introduces a t^n that the dominator must carry. The build caught it, but the magnitude computation should have caught it first.
- **A constant-factor generalisation plus a numerical check is a legitimate proceed-without-GPT case.** The deliberation overhead is best spent where the statement form is genuinely uncertain, not where the proof is a verified template with confirmed closed forms.

7 Follow-ups

- **Moment normalisation.** $Z^{(n)}(0)/Z(0) = \langle (-tx)^n \rangle_0$ ties these derivatives to the raw moments under the unperturbed Gibbs measure; a short corollary once divided through by $Z(0) > 0$.
- **Flow-equation / fourth-cumulant identity.** The motivating application: assemble $Z''(0)$, $Z'''(0)$ (and $Z^{(4)}(0)$, a further $n = 3$ instance) into the connected fourth cumulant. Now unblocked.
- **General- h version.** Centring the dominated-differentiation ball at an arbitrary h_0 rather than 0 would upgrade the step lemma to a genuine “ Z is smooth in h ” statement and allow true iterated `iteratedDeriv` chaining; deferred as the $h = 0$ form is what the cumulant application needs.

²`Continuous.astronglyMeasurable` composed with `fun_prop`.

³`hasDerivAt.integral_of_dominated_loc_of_deriv.le`.